Adaptive-Intent Prediction in Occluded Urban Scenarios

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***Abstract*—Pedestrian crossing intention prediction becomes particularly difficult in urban environments when pedestrians are only partially visible because of parked vehicles, moving traffic, or other roadside obstacles. This work presents an occlusion aware framework for predicting pedestrian crossing intention from incomplete visual observations. Pedestrians are automatically detected and tracked over time, while a learned occlusion estimator represents their visibility using probabilistic evidence. Appearance, spatial, motion, and Bayesian observation reliability information are combined into a 525-dimensional representation over a 30 frame temporal window. A Transformer based model estimates the likelihood of crossing, while Monte Carlo dropout and temperature scaling are used to capture prediction uncertainty and improve confidence calibration. Rather than forcing a binary decision for every observation, a learned selective decision policy considers intent, uncertainty, reliability, and motion information to either commit to crossing, commit to not crossing, or request further evidence. Integrated Gradients are used to provide evidence based explanations for the selected action. Evaluation on held out pedestrian sequences from the PIE dataset achieved an intent prediction accuracy of 87.59%. The learned decision policy achieved 88.94% committed accuracy at 87.15% coverage. An end-to-end runtime implementation further demonstrated operation directly from raw video without requiring ground truth pedestrian bounding boxes or occlusion labels during inference.**

***Index Terms*—Pedestrian intent prediction, occlusion-aware perception, Transformer, uncertainty estimation, selective decision making, autonomous driving.**

I. Introduction

Autonomous vehicles operating in urban environments need to do more than detect pedestrians. They also need to anticipate how a pedestrian is likely to behave, particularly when that person may enter the vehicle's path. An early estimate of crossing intention can provide useful information to downstream planning and safety systems before the crossing action becomes obvious. This remains a difficult problem because pedestrian behavior is influenced by motion, body appearance, position, surrounding traffic, and changes that occur over time. The difficulty becomes greater when some of these cues cannot be observed clearly.

Partial occlusion is common in urban traffic. A pedestrian may appear from behind a parked vehicle, be partly hidden by another road user, or remain visible only through a small region between roadside objects. In such situations, the perception system does not receive the same amount of information at every moment. A pedestrian who is only partly visible in one frame may become clearer several frames later as the scene evolves. Therefore, pedestrian intent prediction under occlusion is not only a question of predicting crossing or not crossing; it is also a question of deciding whether the evidence currently available is sufficient to support that prediction.

Existing pedestrian crossing intention methods have increasingly used temporal and multimodal information, including pedestrian appearance, bounding box motion, pose, vehicle state, and contextual observations. Recent attention and Transformer based approaches have further improved the ability to model dependencies across observation sequences. However, reduced visibility can directly weaken the cues on which these models depend. Occlusion may remove useful appearance information, distort motion estimates, or leave only

a small portion of the pedestrian observable. More importantly, a conventional classifier is generally expected to return a prediction for every input, even when the current observation is ambiguous. A high classification score alone does not necessarily indicate that the model's confidence remains reliable when the available visual evidence has been degraded.

This work addresses the problem from the perspective of adaptive decision making under partial evidence. The proposed framework is developed using real world pedestrian sequences from the Pedestrian Intention Estimation (PIE) dataset. Controlled synthetic occlusion is applied to the pedestrian observations so that the effect of different visibility conditions can be studied systematically while retaining the original temporal behaviour of the recorded sequences. Appearance, spatial, and motion information are represented over a sequence of observations and processed by a temporal intent prediction model to estimate pedestrian crossing behaviour.

II. Related Work

A. Pedestrian Crossing Intention Prediction

Early pedestrian intention studies focused on behavioural cues that could be observed directly from video, such as head orientation, walking motion, and the surrounding traffic context. Varytimidis et al. [1], for example, used visual feature extraction together with conventional machine learning methods to estimate pedestrian motion and head orientation before predicting whether a pedestrian was likely to cross. The introduction of the Pedestrian Intention Estimation (PIE) dataset by Rasouli et al. [2] later provided a larger benchmark for studying pedestrian behaviour from an ego vehicle perspective and supported the development of temporal intention prediction methods.

More recent approaches have increasingly relied on deep temporal and multi modal representations. Yang et al. [3] combined visual, motion, and contextual information using feature fusion and spatio temporal attention, while Cadena et al. [4] incorporated pose, image context, and ego vehicle information through a graph based architecture. Pedestrian vehicle interaction has also become an important source of information. Xu et al. [5] explicitly modelled spatial and temporal relationships between the pedestrian and the ego vehicle, whereas Chen et al. [6] proposed a progressive multimodal token fusion framework that combines bounding boxes, pedestrian pose, and ego vehicle speed. Collectively, these studies show that pedestrian intention is better represented through complementary temporal and contextual cues than through appearance alone.

B. Prediction under Occlusion and Uncertainty

A major difficulty arises when these cues are only partially available. In urban scenes, pedestrians can be hidden by parked vehicles, moving traffic, or roadside objects, reducing the quality of appearance, pose, and motion information. Li et al. [7] considered occlusion as part of complex-scene pedestrian intention recognition and used adaptive fusion of micro-motion and contextual features to improve robustness under varying visibility. Liu et al. [8] addressed the problem more directly through an occlusion-aware diffusion model that reconstructs missing motion patterns before performing intention prediction.

Reduced visibility also makes prediction confidence more important. Zhang et al. [9] proposed TrEP, which combines Transformer based temporal modeling with evidential learning to estimate both pedestrian intention and predictive uncertainty. Chen et al. [10] similarly incorporated uncertainty-aware learning into a cross modal Transformer framework. Beyond pedestrian intention prediction, Guo et al. [11] showed that modern neural networks can produce poorly calibrated confidence estimates, motivating the use of calibration techniques when probability values are used for downstream decisions.

C. Selective Decision-Making from Partial Evidence

Estimating uncertainty is useful, but it does not automatically determine whether a prediction should be accepted when the available evidence is weak. Selective prediction addresses this issue by allowing a model to abstain rather than being forced to produce a committed prediction for every input. Geifman and El-Yaniv [12] demonstrated this idea through Selective Net, where prediction and rejection are learned together.

The present work extends this principle to temporally evolving pedestrian observations. Instead of treating every observation window as equally informative, a calibrated intent predictor is coupled with a learned supervised selective decision policy. The policy can commit to a CROSSING or NOT CROSSING interpretation when the available evidence is sufficient, or return OBSERVE MORE and defer the decision to a later observation window. This links pedestrian intent prediction, uncertainty, and temporal evidence accumulation within a single decision process.

III. Methodology

This section presents the proposed framework for pedestrian crossing intention prediction under partial occlusion. The method is designed to operate on temporally evolving pedestrian observations collected from an ego vehicle perspective. Rather than relying on a single image or a fully visible pedestrian, the framework uses a sequence of observations and learns to make a decision from incomplete and changing visual evidence. The overall pipeline consists of four main stages: pedestrian observation preparation, feature representation, temporal intent prediction, and selective decision making.

A. Overall Framework

The proposed framework is designed to predict pedestrian crossing intention when the pedestrian is only partially observable in an urban scene. Instead of relying solely on a fully visible pedestrian appearance, the system combines visual, spatial, motion, and observation-reliability information over time. This allows the prediction process to continue when the available visual evidence becomes weak due to occlusion.

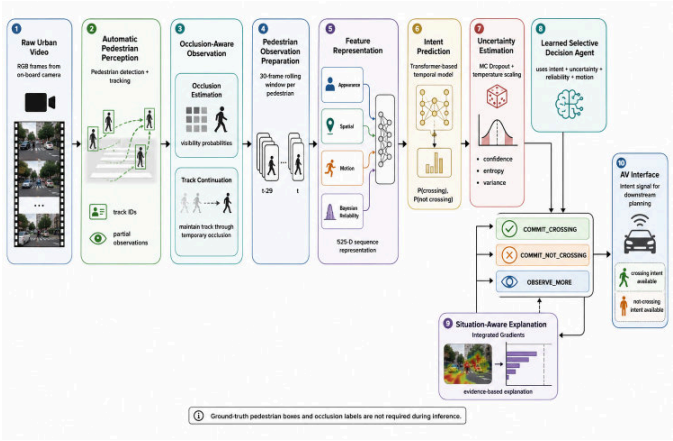


Fig. 1. System Overview

Given an input video, pedestrians are first detected and associated across consecutive frames. The visibility of each pedestrian is then estimated, and the resulting evidence is incorporated into an occlusion-aware feature representation. A fixed temporal window of these features is processed by a Transformer-based intent predictor to estimate the probability of crossing. Prediction uncertainty is subsequently estimated and provided, together with the intent and observation reliability information, to a learned selective decision agent. The agent either commits to a crossing decision, commits to a non-crossing decision, or requests further observation.

The overall process can be expressed as:

$$V \rightarrow T_t \rightarrow O_t \rightarrow X_t \rightarrow P(C_t) \rightarrow U_t \rightarrow A_t, \tag{1}$$

where V denotes the input video, T_t the pedestrian track at time t, O_t the estimated occlusion evidence, X_t the temporal pedestrian representation, $P(C_t)$ the crossing probability, U_t the uncertainty information, and A_t the final agent action.

B. Pedestrian Observation Preparation and Perception

The PIE dataset was used to construct pedestrian observation sequences for model development. Pedestrian samples labeled as crossing and not crossing were retained, while crossing-irrelevant observations were excluded. Continuous observations were divided into fixed length sequences of 30 frames to preserve short-term changes in pedestrian appearance and movement.

To expose the model to incomplete observations during training, synthetic occlusions were applied to pedestrian regions at different visibility levels. This was particularly important because, in practical road scenes, a pedestrian may first appear only around the edge of a parked or moving vehicle.

For the end-to-end runtime system, pedestrian bounding boxes are obtained automatically rather than from ground-truth annotations. A YOLO based pedestrian detector was fine tuned using visible and partially occluded pedestrian samples, and BoT-SORT was used to associate detections over time. When a previously detected pedestrian temporarily disappears behind an obstacle, a short term track continuation mechanism propagates the existing pedestrian hypothesis using recent motion information. This continuation is only applied to previously observed pedestrians and is not intended to infer pedestrians that have never been visible.

C. Occlusion Aware Feature Representation

For each active pedestrian track, four complementary forms of information are extracted: appearance, spatial layout, motion, and observation reliability. Appearance information is represented using a ResNet-18 feature extractor, yielding

$$a_t \in \mathbb{R}^{512}. \tag{2}$$

The spatial descriptor captures the normalized pedestrian center coordinates, width, height, area, and aspect ratio, defined as

$$s_t = [c_x, c_y, w, h, A, r] \in \mathbb{R}^6. \tag{3}$$

Pedestrian motion between consecutive observations is represented by

$$m_t = [\Delta x, \Delta y, v, \theta] \in \mathbb{R}^4, \tag{4}$$

where Δx and Δy describe displacement, v represents motion magnitude, and θ represents direction.

Occlusion is estimated using a learned three-class model that produces the probability distribution

$$q_t = [P(o = N), P(o = P), P(o = F)], \tag{5}$$

where N, P, and F denote no, partial, and full occlusion, respectively. Instead of converting this distribution into a hard visibility label, the probabilities are propagated through a Bayesian observation-reliability model. The final reliability representation is obtained by marginalizing across the possible occlusion states:

$$r_t = \sum_{o \in \{N,P,F\}} P(o \mid I_t)\, r_t^{(o)}, \tag{6}$$

where

$$r_t = [P(R = L), P(R = M), P(R = H)] \in \mathbb{R}^3 \tag{7}$$

represents low, medium, and high observation reliability.

The final pedestrian representation at time step t is formed by concatenating all feature descriptors:

$$x_t = [a_t\,;\, s_t\,;\, m_t\,;\, r_t] \in \mathbb{R}^{525}. \tag{8}$$

A temporal input sequence is constructed using 30 consecutive observations,

$$X_t = [x_{t-29}, \ldots, x_t] \in \mathbb{R}^{30 \times 525}. \tag{9}$$

This probabilistic formulation explicitly preserves uncertainty regarding pedestrian visibility rather than forcing every observation into a single discrete occlusion state.

D. Temporal Intent Prediction

The temporal feature sequence X_t is processed by a Transformer based binary intent classifier. Self attention allows the model to capture relationships and changes across appearance, location, motion, and observation reliability throughout the full observation window. The model outputs a pedestrian crossing probability:

$$p_t = P(C_t = 1 \mid X_t). \tag{10}$$

A calibrated threshold is then applied to obtain the discrete crossing or non crossing prediction.

Since a high crossing probability does not necessarily imply a reliable prediction under severe occlusion, predictive uncertainty is estimated during inference. Monte Carlo Dropout is employed to perform K stochastic forward passes, from which the mean crossing probability is computed as

$$\bar{p}_t = \frac{1}{K} \sum_{k=1}^{K} p_t^{(k)}. \tag{11}$$

The corresponding predictive variance is given by

$$\sigma_t^2 = \frac{1}{K} \sum_{k=1}^{K} \left(p_t^{(k)} - \bar{p}_t \right)^2. \tag{12}$$

Predictive entropy and mutual information are additionally computed to quantify epistemic and aleatoric uncertainties, while temperature scaling is applied to optimize probability calibration. These uncertainty estimates are preserved for the downstream decision making stage rather than forcing every intent prediction into an immediate definitive action.

E. Learned Agentic Decision and Explanation

The final decision stage is implemented as a learned supervised selective policy, rather than a manually defined rule set. The agent receives a 15-dimensional state $z_t \in \mathbb{R}^{15}$ containing the crossing probability, prediction confidence, predictive entropy, mutual information, probability variance, variation ratio, decision margin uncertainty, temporal observation reliability, and pedestrian motion information.

The policy is represented by a multilayer perceptron (MLP):

$$\pi_\theta(z_t) = \mathrm{softmax}\left(f_\theta(z_t)\right). \tag{13}$$

The action space $\mathcal{A}$ comprises three discrete operational states:

$$\mathcal{A} = \{a_{\text{obs}}, a_{\text{not}}, a_{\text{cross}}\}, \tag{14}$$

where a_{obs}, a_{not}, and a_{cross} correspond to `OBSERVE_MORE`, `COMMIT_NOT_CROSSING`, and `COMMIT_CROSSING`, respectively. The selected action a_t^* is determined by:

$$a_t^* = \arg\max_{a \in \mathcal{A}} \pi_\theta(a \mid z_t). \tag{15}$$

This selective formulation allows the system to defer a decision when the available visual evidence is insufficient instead of forcing a crossing or non-crossing prediction in every frame.

For interpretability, the decision agent is coupled with a situation aware explanation module based on Integrated Gradients. The contribution of each agent state feature is measured relative to the selected action, and related features are grouped into interpretable evidence categories such as intent evidence, uncertainty, observation reliability, and motion. The explanation therefore indicates which observed factors most strongly supported or opposed the selected decision. These explanations describe the learned evidence used by the policy and are not interpreted as causal guaranties.

The selected agent action is finally converted into an interface signal that can be consumed by a downstream autonomous-driving system:

$$\begin{cases} \texttt{COMMIT_CROSSING} & \rightarrow \texttt{CROSSING_INTENT_AVAILABLE}, \\ \texttt{COMMIT_NOT_CROSSING} & \rightarrow \texttt{NOT_CROSSING_INTENT_AVAILABLE}, \\ \texttt{OBSERVE_MORE} & \rightarrow \texttt{MORE_EVIDENCE_REQUIRED}. \end{cases} \tag{16}$$

IV. Experimental Setup and Evaluation

A. Dataset and Evaluation Protocol

Experiments were conducted using the Pedestrian Intention Estimation (PIE) dataset, which contains pedestrian trajectories captured in real urban traffic scenes together with behavioral and contextual annotations. Pedestrian observations labeled crossing and not crossing were used, while crossing irrelevant samples were excluded. Each pedestrian trajectory was divided into fixed observation windows of 30 frames, consistent with the temporal input required by the proposed model.

To reduce identity leakage between training and evaluation, the dataset was divided using pedestrian level separation so that the pedestrians used for testing were not included in the training set. The resulting test set contained 1,152 pedestrian sequences. Synthetic occlusion was applied during training to expose the model to different levels of visibility degradation, while the final runtime system estimated pedestrian occlusion automatically.

B. Implementation Details

The pedestrian representation consisted of a 512-dimensional ResNet-18 appearance embedding, six spatial features, four motion features, and three Bayesian observation reliability probabilities, producing a 525-dimensional feature vector per frame. Thirty consecutive feature vectors were provided to the Transformer-based intent predictor.

Prediction uncertainty was estimated using Monte Carlo dropout with 30 stochastic forward passes. Temperature scaling was used for probability calibration, and the calibrated uncertainty information was subsequently provided to the learned selective decision policy.

For end to end video processing, a fine tuned YOLO based pedestrian detector was used together with BoT-SORT for temporal association. A learned three class occlusion estimator predicted the probability of no, partial, or full occlusion. Temporary loss of a previously established track was handled using an occlusion aware continuation mechanism. The final runtime therefore operated directly on RGB video without

requiring ground truth pedestrian bounding boxes, pedestrian identities, or occlusion labels as inference inputs.

C. Evaluation Metrics

Intent prediction performance was evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and average precision (AP). Since the final decision agent can defer uncertain observations, its performance was additionally measured using committed accuracy and coverage. Coverage represents the proportion of observations for which the learned agent commits to either crossing or not crossing, while the remaining observations are assigned to the observe more action.

The automatic perception components were evaluated separately using standard detection and tracking measures, while qualitative end-to-end video tests were used to verify that the complete pipeline could generate pedestrian intent decisions from raw visual input under partial and occluded observations.

V. Results and Discussion

A. Intent Prediction Performance

The final Transformer based intent model was evaluated on 1,152 held-out pedestrian sequences. The model achieved an overall accuracy of 87.59%, with a ROC-AUC of 88.28%, indicating that the temporal representation was able to separate crossing and non-crossing behavior reasonably well under varying observation quality.

TABLE I
Intent Prediction Performance

Metric	Result
Accuracy	87.59%
Precision	61.63%
Recall	75.50%
F1-score	67.87%
ROC-AUC	88.28%
Average Precision	69.28%

B. Selective Decision Performance

The learned decision agent was evaluated using the calibrated intent and uncertainty outputs together with observation reliability and motion information. Unlike the underlying intent classifier, the agent was allowed to defer uncertain observations through the OBSERVE MORE action.

The learned policy committed to a final crossing or not-crossing decision for 87.15% of the test observations and achieved a committed accuracy of 88.94%. The remaining 12.85% of observations were deferred for additional evidence.

TABLE II
Selective Decision Performance

Measure	Result
Base intent accuracy	87.59%
Decision coverage	87.15%
Committed accuracy	88.94%
Deferred observations	12.85%

Although the improvement in committed accuracy is moderate, the result is meaningful because it was obtained without a manually defined rule based decision layer. Instead, the policy learned when the available combination of intent probability, uncertainty, reliability, and motion evidence was sufficient to support a commitment. This selective behavior is particularly useful under occlusion, where forcing a binary decision from weak visual evidence may reduce reliability.

C. Automatic Perception and Occlusion Analysis

The final system was also evaluated with automatically generated pedestrian observations rather than ground truth bounding boxes. The fine tuned YOLO pedestrian detector achieved a precision of approximately 43.3%, recall of 38.1%, mAP@0.5 of 30.6%, and mAP@0.5:0.95 of 19.9% on the video-disjoint detector validation split.

The learned occlusion estimator achieved 58.0% classification accuracy and a macro-F1 score of approximately 57.8% over the three visibility states: none, partial, and full occlusion. These results indicate that automatic visibility estimation remains substantially more difficult than intent prediction itself. For this reason, the system does not propagate the estimated occlusion as a hard label. Instead, the complete probability distribution is marginalized through the Bayesian observation reliability model before being passed to the temporal intent predictor.

Tracking was also found to be the most challenging component under severe occlusion. Short term track continuation improved temporal coverage when an already observed pedestrian became temporarily hidden; however, it does not attempt to infer a pedestrian that has never been visible. This represents an important practical limitation of a monocular RGB based system.

D. End-to-End Runtime Analysis

The complete pipeline was finally tested directly on raw video. In this configuration, pedestrian detection, tracking, occlusion estimation, reliability modelling, intent prediction, uncertainty estimation, selective decision-making, and explanation were executed without supplying ground-truth pedestrian bounding boxes, pedestrian identities, occlusion labels, or crossing labels to the runtime system.

In one representative occluded sequence, the automatic pipeline processed 270 frames and generated 30 complete intent decisions from automatically established pedestrian tracks. The runtime also demonstrated that a previously observed pedestrian hypothesis could be maintained during temporary visual disappearance and subsequently processed using the same 30-frame intent model.

A representative output is shown in Fig. 1, where a pedestrian is largely hidden by a vehicle. The system identifies the observation as strongly occluded, assigns low observation reliability, and propagates this information to the learned decision policy before producing the final intent signal.

These end-to-end experiments demonstrate the feasibility of the proposed architecture, but they should not be interpreted as a separate end-to-end accuracy benchmark. The quantitative intent results in Table I were obtained from the held out

pedestrian test sequences, while detector, occlusion, and tracking components were evaluated independently. A larger fully annotated end-to-end video benchmark remains an important direction for future evaluation.

VI. Conclusion

This work presented an occlusion aware framework for pedestrian crossing intention prediction in urban scenes where visual evidence may be incomplete. The proposed system combines automatic pedestrian detection and tracking, probabilistic occlusion estimation, Bayesian observation reliability modeling, Transformer based temporal intent prediction, uncertainty estimation, and a learned selective decision policy. The intent predictor achieved an accuracy of 87.59% on held out pedestrian sequences from the PIE dataset, while the learned decision policy achieved 88.94% committed accuracy at 87.15% coverage. The end-to-end runtime further demonstrated that crossing intention decisions can be generated directly from raw video without requiring ground truth pedestrian bounding boxes or occlusion labels during inference. The selective decision mechanism also enables the system to request additional evidence when the available observation is uncertain rather than forcing an immediate binary decision. Future work will focus on improving pedestrian detection and track continuity under severe occlusion, strengthening automatic occlusion estimation, and evaluating the framework on a wider range of real world and simulated urban scenarios.

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